

MIAMI OPA LOCKA, FL, USA

WMO: 722024

Lat: 25.907N

Lon: 80.280W

Elev: 10

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 12882

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	47.5	51.6	31.8	26.1	55.3	36.6	31.8	56.7	24.3	72.1	21.4	72.2	8.5	340	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.7	92.2	78.1	91.1	78.0	90.3	77.9	81.2	86.8	80.3	86.3	79.7	86.0	10.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.5	153.1	83.2	79.0	150.7	83.1	78.1	146.1	82.8	44.7	86.5	43.7	85.9	43.0	85.2	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.8	18.9	17.4	41.1	94.8	3.7	1.6	38.4	95.9	36.3	96.9	34.2	97.8	31.5	98.9		
			36.7	82.0	3.9	1.0	33.9	82.7	31.6	83.3	29.3	83.8	26.5	84.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	76.9	67.3	70.2	72.4	76.0	79.5	82.5	83.5	84.0	82.7	79.7	73.7	70.8
	DBStd	7.11	6.94	5.93	5.41	3.90	2.70	2.33	2.04	1.93	2.10	3.56	4.71	6.31
	HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD65	127	60	24	11	0	0	0	0	0	0	0	5	27
	CDD50	9814	537	565	694	780	915	974	1038	1054	980	921	712	644
	CDD65	4466	130	169	240	331	450	524	573	589	530	457	267	206
	CDH74	40951	580	872	1466	2425	4042	5545	6546	6845	5698	4241	1643	1048
CDH80	13419	41	123	295	599	1238	2060	2642	2814	2042	1194	257	114	
Wind	WSAvg	8.3	8.7	8.7	9.6	9.6	9.1	7.3	6.6	6.6	7.2	8.7	8.6	8.5
Precipitation	PrecAvg	60.1	1.9	2.4	2.4	3.0	4.9	9.0	7.2	8.2	9.5	6.6	2.7	2.2
	PrecMax	75.3	6.9	5.9	6.4	6.1	10.6	17.3	11.3	13.1	15.3	15.2	9.9	10.0
	PrecMin	50.0	0.1	0.2	0.2	0.4	0.0	2.8	4.0	2.9	5.4	1.4	0.4	0.2
	PrecStd	7.5	1.6	1.5	1.5	1.7	2.7	4.0	2.0	2.5	3.0	4.1	2.5	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.2	85.9	88.5	90.1	91.3	93.3	93.2	93.3	92.3	90.4	87.1	85.1
		MCWB	72.0	72.4	74.4	73.5	74.7	77.8	78.5	78.6	78.4	77.3	75.2	73.9
	2%	DB	81.3	83.2	85.7	87.7	89.4	91.2	91.5	92.0	90.7	88.5	84.6	82.6
		MCWB	71.1	71.9	73.1	73.2	74.8	77.7	78.3	78.6	78.3	76.9	74.5	73.3
	5%	DB	79.5	81.4	83.3	85.5	87.9	90.1	90.5	91.0	89.7	87.5	82.6	81.2
		MCWB	69.8	70.8	72.0	72.4	74.6	77.6	78.2	78.5	78.2	76.6	72.9	72.2
	10%	DB	77.7	79.5	81.2	83.6	86.1	88.5	89.7	90.0	88.3	85.8	81.3	79.4
		MCWB	68.6	69.6	70.7	71.5	73.9	77.1	78.2	78.5	77.7	75.7	71.8	71.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.6	75.3	78.9	76.5	79.1	80.7	83.5	82.1	82.8	80.4	77.8	76.9
		MCDB	79.2	81.2	83.2	85.2	85.3	87.7	88.0	87.9	87.6	84.7	82.5	80.5
	2%	WB	73.1	73.7	75.5	75.4	77.8	79.7	80.9	80.8	81.0	79.3	76.4	75.4
		MCDB	78.1	79.4	81.7	82.6	84.9	87.0	86.4	87.6	85.7	83.9	81.5	80.1
	5%	WB	71.9	72.7	74.1	74.5	76.8	78.9	79.9	80.1	80.0	78.4	75.5	74.5
		MCDB	77.1	78.5	80.0	81.6	83.9	86.1	86.4	87.0	85.2	83.5	80.6	79.1
	10%	WB	70.5	71.5	72.7	73.4	75.8	78.2	79.1	79.4	79.1	77.6	74.3	73.2
		MCDB	76.0	77.4	78.7	80.4	82.7	85.4	86.0	86.3	84.7	83.2	79.2	77.8
Mean Daily Temperature Range	5% DB	MDBR	16.4	16.0	15.7	14.8	13.6	12.9	12.9	12.7	12.0	11.9	14.0	14.5
		MCDBR	15.2	15.8	16.1	16.2	15.3	14.2	13.7	13.5	13.0	12.6	13.7	14.0
		MCWBR	7.6	6.8	6.6	6.4	5.6	5.3	5.4	4.9	4.7	4.6	5.9	6.9
	5% WB	MCDBR	13.6	14.4	14.7	14.7	13.7	13.2	13.1	12.7	11.9	11.5	12.3	12.1
		MCWBR	8.1	7.0	6.9	6.5	5.6	5.6	5.9	5.3	5.2	5.0	5.8	7.2
Clear-Sky Solar Irradiance	taub		0.364	0.373	0.386	0.400	0.427	0.461	0.466	0.458	0.434	0.415	0.381	0.373
	taud		2.522	2.512	2.461	2.408	2.375	2.337	2.317	2.366	2.433	2.455	2.498	2.514
	Ebn at Noon		282	288	289	285	276	265	263	266	271	270	274	273
	Edh at Noon		30	32	35	38	39	41	41	39	36	33	30	29
All-Sky Solar Radiation	RadAvg		1166	1414	1727	1959	1970	1796	1840	1773	1562	1440	1234	1080
	RadStd		76	82	84	92	161	159	84	95	78	101	58	79

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.99	+1.89	N/A	+0.61	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)	+0.68	+1.55	N/A	N/A	N/A	N/A	N/A	-38	+234	+184

Nomenclature: See separate page